

Complete exceptions to a distance Laplacian inequality for graphs of maximum degree $n - 2$

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Abstract

Let $U_r(G)$ be the sum of the r largest distance Laplacian eigenvalues of a connected simple graph G , and let $W(G)$ be its Wiener index. For graphs of order $n \geq 3$ and maximum degree exactly $n - 2$, we classify all pairs (G, r) satisfying $U_r(G) > W(G) + \binom{r+2}{3}$. There are precisely thirteen pairs on eleven graph isomorphism classes, all of order at most seven. Consequently the least eventual threshold is $N = 8$, refining the previously established sufficient threshold 10. The proof combines an elementary argument for every $n \geq 10$ with exact certification of all 67 501 eligible graphs through order nine. The finite verification covers all 602 945 pairs and uses integer inequalities and rational root certificates, with no floating-point decisions.

1 Statement and context

The graph G is finite, simple, undirected and connected; auxiliary graphs need not be connected. For G on n vertices, write $d(u, v)$ for its shortest-path distance and define

$$D = (d(u, v))_{u, v}, \quad L_D(G) = \text{diag}\left(\sum_v d(u, v)\right)_u - D, \quad W(G) = \sum_{u < v} d(u, v).$$

Let $\lambda_1 \geq \dots \geq \lambda_n = 0$ be the eigenvalues of $L_D(G)$, counted with multiplicity, and put $U_r(G) = \sum_{i=1}^r \lambda_i$ for $1 \leq r \leq n$. We study the distance Laplacian inequality

$$U_r(G) \leq W(G) + \binom{r+2}{3} \tag{1}$$

under the equality condition $\Delta(G) = n - 2$ on the maximum degree.

Huang [1, Theorem 5.2] proves that (1) holds in this family for every $n \geq 10$ and every $1 \leq r \leq n$. Mushtaq, Pirzada and Khan [2, Theorem 2.5] give the earlier sufficient condition $n \geq 95$; this order hypothesis occurs in the theorem statement. The result below completes the exceptional small orders and determines the least possible eventual threshold.

Theorem 1. *Let G have order $n \geq 3$ and maximum degree exactly $n - 2$. For an integer $1 \leq r \leq n$, the strict reverse of (1) holds if and only if (G, r) is isomorphic to a pair listed in Table 1, with r retained as part of the pair. There are thirteen such pairs on eleven graph isomorphism classes. The least integer $N \geq 3$ such that (1) holds for every eligible graph of every order $n \geq N$ and every $1 \leq r \leq n$ is $N = 8$.*

Table 1: Complete strict-violation pairs. Each edge ij means $\{i, j\}$; the vertex set is $\{0, \dots, n-1\}$. The graph6 code identifies the same representative.

n	graph6	W	Violating r	Edge set
4	CR	10	2	02, 13, 23
5	D@s	18	2, 3	23, 04, 14, 34
5	DBw	16	3	13, 23, 04, 14, 24
5	DD[	16	3	03, 23, 14, 24, 34
5	D' [	17	3	01, 23, 14, 24, 34
6	E?Fg	28	3, 4	34, 05, 15, 25, 45
6	E?NW	26	3	24, 34, 05, 15, 35, 45
6	EGEw	27	3	12, 34, 05, 25, 35, 45
6	EAMw	25	3	13, 24, 34, 05, 25, 35, 45
6	E@NW	25	3	23, 24, 34, 05, 15, 35, 45
7	F??Ng	40	4	45, 06, 16, 26, 36, 56

The classification includes non-trees and every value of r , including $r = n$. Equality in (1) is not an exception. The infinite-order argument follows the distance-layer approach in [1]; we give its specialization in full, including a discrete check of the remaining five values of r . The contribution relative to the two inspected primary texts is the complete exception table and the sharp threshold, rather than the already known large-order inequality or the seven-vertex lower-bound example. This comparison is limited to those texts and makes no exhaustive priority claim. For adjacent graph-spectral computation by integer characteristic polynomials and Sturm root counts, see Zeng [3].

2 Distance layers and the infinite tail

Choose a vertex v of degree $n-2$ and let x be its unique nonneighbor. Every two vertices of $C = N[v]$ have distance at most two. Since x is the only vertex outside C , connectedness gives it a neighbor in C . Thus $\text{diam}(G) \leq 3$, and every distance-three pair contains x .

For $s = 2, 3$, define a graph Y_s on $V(G)$ by declaring uw an edge if and only if $d(u, w) \geq s$. Put $m = |E(Y_2)|$ and $t = |E(Y_3)|$. The graph Y_3 is a star with t edges together with isolated vertices, or is edgeless if $t = 0$. Comparing off-diagonal entries and row sums gives

$$L_D = nI - J + L(Y_2) + L(Y_3), \quad W = \binom{n}{2} + m + t, \quad (2)$$

where L denotes the ordinary graph Laplacian and J the all-ones matrix. Also

$$z^T L_D z = \sum_{u < w} d(u, w) (z_u - z_w)^2. \quad (3)$$

Consequently L_D has exactly $n-1$ positive eigenvalues, its kernel consists of constant vectors, and $\text{tr}(L_D) = 2W$.

Write $S_r(H)$ for the sum of the r largest ordinary Laplacian eigenvalues of an n -vertex graph H with e edges. These eigenvalues are nonnegative, sum to $2e$, and are at most n . Indeed, $L(H) + L(\overline{H}) = nI - J$, and positivity of $L(\overline{H})$ gives the last assertion on the orthogonal complement of the constants. Therefore

$$S_r(H) \leq \min(2e, nr) \leq e + \frac{nr}{2}. \quad (4)$$

For $t > 0$, the nonzero spectrum of the star with t edges is $t + 1$ and $t - 1$ copies of 1. Hence

$$S_r(Y_3) = t + \min(r, t) \leq t + r \quad (t > 0), \quad S_r(Y_3) = 0 \quad (t = 0). \quad (5)$$

For a real symmetric matrix A , the sum of its r largest eigenvalues is the maximum of $\sum_{i=1}^r z_i^\top A z_i$ over orthonormal r -frames. To see this, expand the frame in an eigenbasis: its squared coordinates give weights in $[0, 1]$ whose sum is r , and the maximum places weight 1 on the largest r eigenvalues. Applying this formula separately to the summands in (2), and using (4)–(5), yields

$$U_r \leq nr + m + \frac{nr}{2} + t + r = W - \binom{n}{2} + \frac{3nr}{2} + r, \quad 1 \leq r \leq n - 1. \quad (6)$$

Lemma 2. *Every graph with $n \geq 10$ and $\Delta(G) = n - 2$ satisfies (1) for every $1 \leq r \leq n$.*

Proof. For $r \leq n - 1$, (6) reduces the assertion to

$$F(n, r) = 3n^2 - 3n + r^3 + 3r^2 - 4r - 9nr \geq 0.$$

For $r \geq 6$, completing the square gives

$$F(n, r) = 3 \left(n - \frac{3r+1}{2} \right)^2 + g(r), \quad g(r) = r^3 - \frac{15}{4}r^2 - \frac{17}{2}r - \frac{3}{4}.$$

Writing $u = r - 6 \geq 0$, we have

$$g(u+6) = u^3 + \frac{57}{4}u^2 + \frac{109}{2}u + \frac{117}{4} > 0.$$

For the integers $1 \leq r \leq 5$, put $z = n - 10 \geq 0$. Then

$$F(n, r) = 3z^2 + (57 - 9r)z + F(10, r), \quad (F(10, r))_{r=1}^5 = (180, 102, 42, 6, 0).$$

All coefficients are nonnegative. This proves the required assertion for $r \leq n - 1$. For $r = n$, use $U_n = 2W$, the diameter bound, and

$$W \leq 3 \binom{n}{2} < \binom{n+2}{3}, \quad \binom{n+2}{3} - 3 \binom{n}{2} = \frac{n((n-3)^2 + 2)}{6} > 0.$$

□

The discrete check above also addresses a local slip in the proof of [1, Theorem 5.2]: the derivative of $r^3 + 3r^2 - 94r + 270$ is asserted there to be negative throughout $[1, 5]$, but it equals 11 at $r = 5$. The five integer values displayed above establish the needed conclusion without that assertion.

3 Complete finite certification

3.1 Generation and elementary certificates

There is no eligible graph of order three: maximum degree one cannot connect three vertices. For each $4 \leq n \leq 9$, the nauty program **geng**, invoked as

$$\text{geng} -c -D\langle n-2 \rangle -1 \langle n \rangle,$$

generates connected graphs of maximum degree at most $n - 2$ once per isomorphism class. We then retain exactly those with maximum degree $n - 2$. The complete graph6 lists are supplied with the paper. Their counts are

n	3	4	5	6	7	8	9
Eligible classes	0	2	8	49	344	3 687	63 411

The completeness of the documented **geng** algorithm is an explicit software dependency. As checks on its use, generation was repeated without degree pruning, followed by the exact degree filter; the resulting sets agreed at every order. All 403 eligible classes through order seven were also matched to the graph atlas by exact isomorphism tests. The two generation runs use the same generator implementation; they are not a formal verification of that algorithm.

Every graph is checked at every $1 \leq r \leq n$. Distances, transmissions, W and all entries of L_D are computed as integers. Put

$$R = W + \binom{r+2}{3}, \quad q = n - 1, \quad s = 2W, \quad Q = \text{tr}(L_D^2).$$

Three sufficient tests certify most nonviolations. First, positivity gives $U_r \leq s$, so $s \leq R$ suffices. Second, if $1 \leq r < q$, Cauchy–Schwarz on the first r and remaining $q - r$ positive eigenvalues gives

$$Q \geq \frac{U_r^2}{r} + \frac{(s - U_r)^2}{q - r}, \quad U_r \leq \frac{rs + \sqrt{r(q - r)(qQ - s^2)}}{q}.$$

Thus the entirely integer certificate

$$A = qR - rs \geq 0, \quad A^2 \geq r(q - r)(qQ - s^2) \quad (7)$$

proves $U_r \leq R$. The quantity $qQ - s^2$ is nonnegative by Cauchy–Schwarz, and is checked as well. In particular, squaring in (7) does not discard a sign condition. Finally, the sharper versions of (4) and (5) give

$$U_r \leq nr + \min(2m, nr) + \begin{cases} t + \min(r, t), & t > 0, \\ 0, & t = 0, \end{cases} \quad 1 \leq r \leq n - 1. \quad (8)$$

The verifier also checks that all distance-three pairs share a vertex. Both orders eight and nine are decided completely by these integer tests.

3.2 Residual algebraic comparisons and verification

Only 49 pairs on 44 graphs, all of order at most seven, remain after the elementary tests. For each such graph the certificate supplies the integer matrix L_D , its integer characteristic polynomial, and disjoint rational intervals isolating all roots with their multiplicities. The intervals are ordered, so the sum of the first r lower and upper endpoints in descending order encloses U_r . A positive lower bound for $U_r - R$ certifies a strict violation; a negative upper bound certifies a strict nonviolation. A singleton interval at zero certifies equality. There is no tolerance-based classification or unresolved interval.

One verifier recomputes the characteristic polynomial by exact Bareiss determinants at $n + 1$ integers. Since the determinant and the supplied polynomial have degree n , these evaluations establish polynomial equality. Square-free factorization and Sturm counts check each isolating interval and multiplicity; singleton roots are evaluated exactly. The multiplicities sum to n , accounting for the whole spectrum. An additional implementation uses Floyd–Warshall distances, exact matrix-power traces and Newton identities for the polynomials, and polynomial arithmetic over rational numbers for the Sturm tests. Neither verifier imports the certificate producer.

The final partition of all pairs is

Certificate or outcome	Number of pairs
Trace bound	258 776
Moment bound	344 110
Distance-layer bound	10
Algebraic strict nonviolation	35
Algebraic equality	1
Algebraic strict violation	13
Total	602 945

The sole equality among the residual algebraic comparisons has graph6 code `EC\w`, $r = 3$ and $W = 24$. Its spectrum is $\{14, 11, 9, 7, 7, 0\}$, giving $U_3 = 34 = W + \binom{5}{3}$. This is not an equality classification for the whole family: other equalities can be certified by the elementary bounds. All thirteen strict violations are precisely the pairs in Table 1.

4 Sharpness and proof of the classification

For the seven-vertex tree in Table 1, direct calculation gives $\Delta = 5$, $W = 40$, and characteristic polynomial

$$p(z) = z(z - 14)^3(z^3 - 38z^2 + 455z - 1722).$$

The cubic factor takes values -12 at 18 and 64 at 19 , so it has a root $\rho \in (18, 19)$. Selecting this eigenvalue and the three copies of 14 yields

$$U_4 \geq \rho + 42 > 60 = W + \binom{6}{3}.$$

This argument does not require knowing the position of ρ in the ordered spectrum. It proves that every proposed threshold $N \leq 7$ fails. The complete finite certificate excludes all violations at orders eight and nine, while Lemma 2 excludes every order at least ten. Thus $N = 8$. The same finite certificate gives exactly Table 1 at all smaller orders, proving Theorem 1 with its full range of r .

Reproducibility and assistance

The accompanying supplement contains the six complete graph6 corpora, the decision ledger, the 44 algebraic certificates, the exception table with rational violation margins, and separate verification code. Its `README.md` gives the synchronous replay commands and software requirements. The main replay uses Python integers, NetworkX for graph parsing and shortest paths, and SymPy for exact polynomial operations; the recorded versions are NetworkX 3.4.2 and SymPy 1.13.3. Generation uses `geng`. The additional rational verifier uses Python’s `Fraction` arithmetic for its spectral checks. The complete finite proof remains conditional on correct execution of these explicitly identified software components, particularly the generator’s completeness contract.

OpenAI Codex was used to assist proof development, code generation, certificate checking and manuscript preparation. Floating-point calculations served only to locate candidates during discovery; all decisions used in this proof were recomputed with integer or rational arithmetic. A separate model-context review reconstructed the argument and checked the finite certificates with a different implementation. Such a review is not external peer review or statistical independence, and no proof-assistant formalization is claimed.

References

- [1] S. Huang, *On a distance Laplacian analog of Brouwer's conjecture for several classes of graphs*, arXiv:2606.06945v1 (2026). <https://arxiv.org/abs/2606.06945v1>.
- [2] U. Mushtaq, S. Pirzada and S. Khan, *On the sum of the eigenvalues of the distance Laplacian matrix of graphs with diameter three and four*, Revista de la Unión Matemática Argentina **69** (2026), 193–202. <https://doi.org/10.33044/revuma.5147>.
- [3] Z. Zeng, *Exact Verification of the Line-Graph Signature Inequality at Order Nine*, SSRN preprint (2026). <https://doi.org/10.2139/ssrn.7386040>.